SHINRA | RETINAL FUNDUS SCREENING SYSTEM

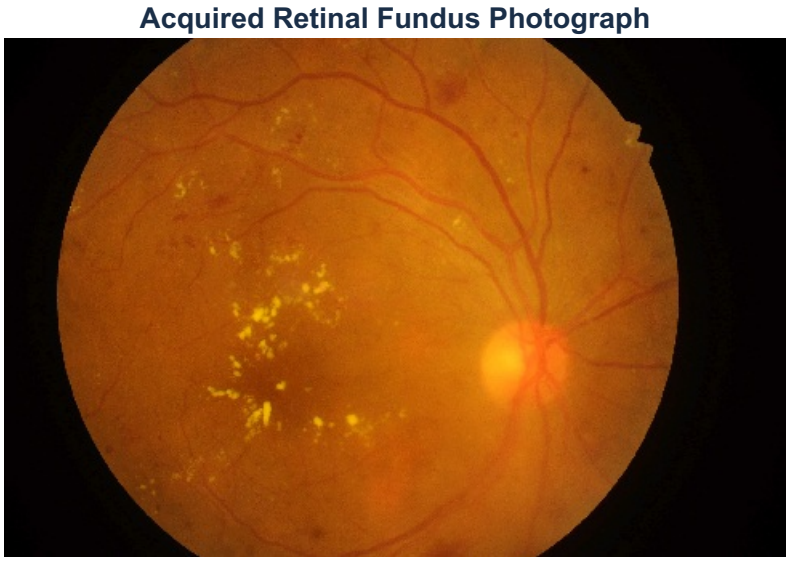
Explainable AI for Diabetic Retinopathy Screening in Rural India

REPORT ID: IDRiD_001

DATE: 2026-10-15 11:09:14

EXECUTIVE CLINICAL SCREENING SUMMARY

Acquired Retinal Fundus Photograph



PATIENT & EXAMINATION METADATA

Patient / Screening ID: IDRiD_001

Examination Timestamp: 2026-10-15 11:09:14

Screening System Engine: SHINRA Multi-Modal AI Engine V2

Software Release: Release V2 (Development Frozen)

Deployment Location: Primary Health Center (Rural Offline Mode)

Input Image Resolution: 512 x 512 pixels

Decision Protocol: Calibrated Safety Gating (tau_ref = 0.30)

Clinical Safety Status: Automated Quality & Reliability Gating Active

1. IMAGE QUALITY SAFETY GATE

Quality Classification: GOOD

Composite Quality Score: 78.4 / 100

Safety Gradability Status: GRADABLE (Proceed with Autonomous Screening)

Operational Guidance: Image quality is clinically acceptable for automated AI screening.

2. DIABETIC RETINOPATHY DIAGNOSTIC FINDINGS

Predicted Severity: Grade 3 — Severe NPDR

Model Confidence: 78.3%

Expected Ordinal Score E[G]: 2.78 (Scale: 0.00 [No DR] to 4.00 [Proliferative DR])

Referable DR Decision: REFERABLE DR

Calibrated Referral Probability: 100.0% (Raw: 100.0% | Operating Threshold tau_ref = 0.30 | Concordance: 95.0%)

3. RELIABILITY SAFETY LAYER & CLINICAL TRIAGE DISPOSITION

RECOMMENDED ACTION: >>> ACCEPT SCREENING <<<

Operational Rationale: High reliability screening (90.0/100). Referable diabetic retinopathy predicted (P_ref = 100.0%).

Composite Reliability Score: 90.0 / 100 (Tier: HIGH | Gate tau >= 60.0: PASSED)

Clinical Urgency: Specialist Ophthalmology Referral

Recommended Window: 2 to 4 weeks (Ophthalmology Clinic Referral)

4. CLINICAL LIMITATIONS & SAFETY NOTICES

- RESEARCH / INVESTIGATIONAL USE ONLY — NOT AN INDEPENDENT MEDICAL DEVICE.
- Designed as an explainable tele-screening aid for rural community clinics in India.
- All algorithmic findings require confirmatory review by a licensed ophthalmologist before treatment.
- Neovascularization and macular edema proximity are automated 2D risk indicators and do not replace OCT.
- **Reviewing Clinician Sign-Off: Name: _____ Date: _____ Signature: _____**

CLINICAL EVIDENCE & RETINAL BIOMARKER INVENTORY

1. RETINAL ANATOMICAL LANDMARKS & GEOMETRIC MORPHOMETRY

• Optic Nerve Head (Optic Disc Landmark):

- Geometric Center Location: (X = 381, Y = 322) pixels
- Disc Radius: 40 pixels
- Localization Confidence: 0.85 / 1.00 (Reliable Landmark)

• Macula & Foveal Risk Zone:

- Estimated Foveal Center: (X = 173, Y = 337) pixels
- Macular Threat Radius (1-DD): 120 pixels (2.5x Optic Disc Radius)
- DME Proximity Risk Factor: Grade 2 / 3 (Lipid exudates within macula)

2. RETINAL VASCULATURE ARCHITECTURE (DEEP U-NET)

• Vessel Segmentation Framework: Deep U-Net Trained Network Architecture

• Total Vascular Caliber Density: 2.26% of total retinal field

• Mean Vascular Tortuosity Index: 1.11 (Reference normal range: 1.05 - 1.25)

• Vascular Bifurcations / Branches: 7 detected vessel branch points

• Clinical Relevance in Diabetic Eye Disease:

- Vessel density attenuation indicates capillary drop-out and retinal non-perfusion.
- Elevated tortuosity index correlates with vascular remodeling and progression toward proliferative DR.

3. QUANTITATIVE RETINAL LESION BIOMARKER INVENTORY

BIOMARKER	QUANTITATIVE MEASUREMENT	CLINICAL SIGNIFICANCE & STATUS
Microaneurysms (MAs)	101 lesions | 1023 px area	Earliest visible sign of NPDR; vascular pouching
Intraretinal Hemorrhages (HEs)	7 lesions | 3.91% area	Dot/blot capillary ruptures; severe NPDR marker
Hard Exudates (EXs)	58 lesions | 1.29% area	Lipid & lipoprotein extravasation from leaky capillaries
Macular DME Threat Grade	Level 2 / 3	Risk of Diabetic Macular Edema threatening central vision
Neovascularization Index (NV)	Biomarker Score: 0.32	Exploratory vessel morphometry (Not clinically certified)

4. METHODOLOGICAL DISCLOSURES & VALIDATION STATUS

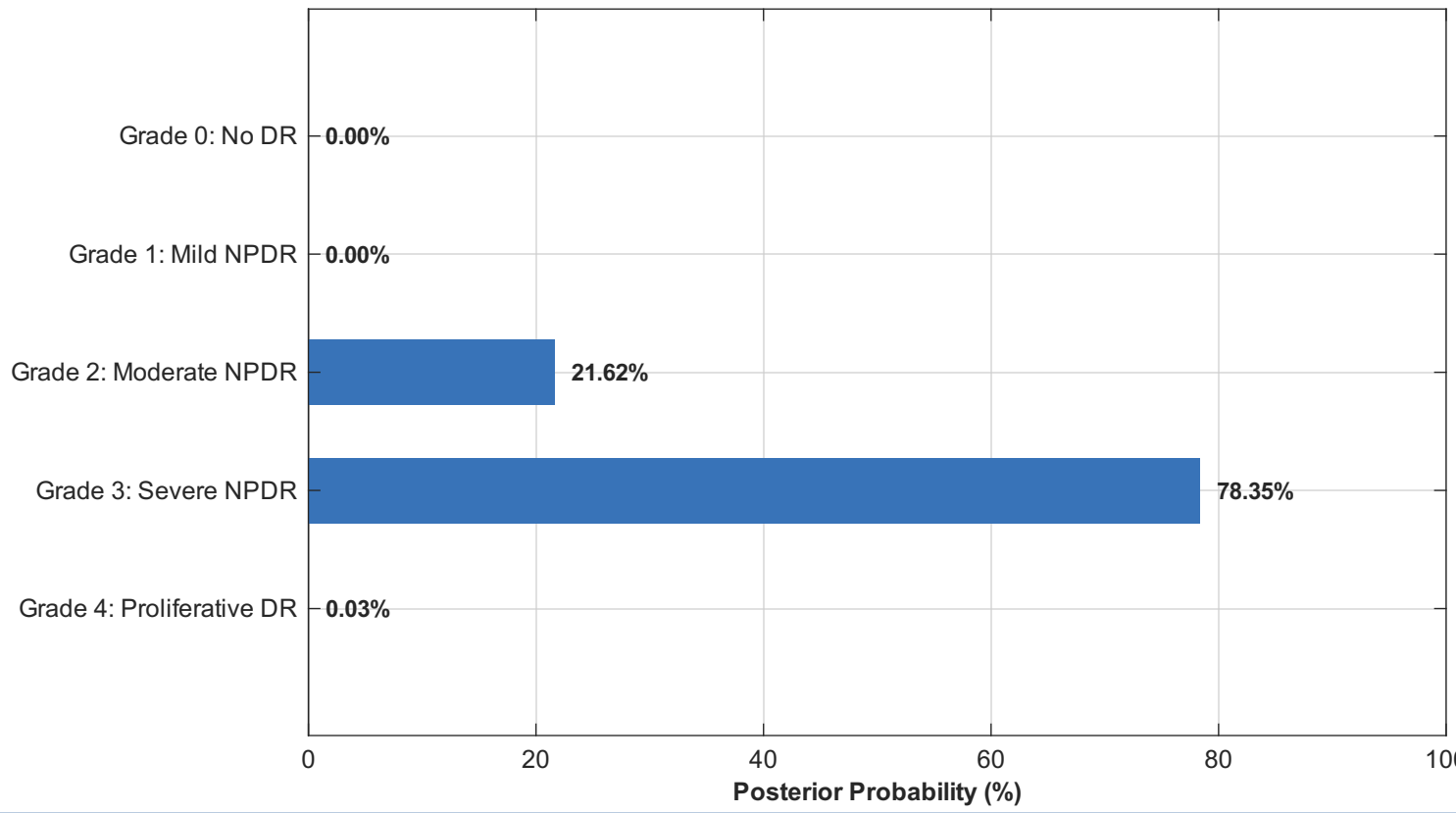
• Microaneurysms, hemorrhages, and exudates are detected using calibrated multi-scale morphological filtering and CLAHE contrast enhancement.

• Neovascularization analysis is an exploratory biomarker derived from vascular branch complexity and does not replace fluorescein angiography.

• DME proximity grading measures Euclidean distance between hard exudates and the estimated fovea; OCT confirmation is required for diagnosis.

MODEL OUTPUTS & 5-FACTOR RELIABILITY ANALYSIS

1. 5-CLASS ICDR DR SEVERITY POSTERIOR PROBABILITY DISTRIBUTION



SEVERITY SUMMARY

Predicted Grade:
Grade 3 — Severe NPDR

Model Confidence:
78.3%

Expected Score E[G]:
2.78 / 4.00

Dominant Class: G3 NPDR

2. REFERABLE DR DECISION & PROBABILITY CALIBRATION

- Raw Referable DR Probability: 100.00% (Sum of Grade 2, 3, 4 posterior probabilities)
- Platt-Calibrated Probability: 100.00% (Fitted on 330-image development partition; a = 1.3106, b = -1.1412)
- Operating Referral Threshold: tau_ref = 0.30 (Frozen development plateau operating point)
- Referral Screening Classification: REFERABLE DR (Calibrated P_ref >= tau_ref)
- Multimodal Concordance Metric: 95.0% agreement between CNN prediction and lesion burden

3. 5-FACTOR ENGINEERING RELIABILITY BREAKDOWN

RELIABILITY COMPONENT	WEIGHT	SCORE [0-100]	FACTOR ASSESSMENT & RATIONALE
1. Image Quality (S_q)	25%	**78.4**	Classification: GOOD (Quality Score: 78.4)
2. Classifier Confidence (S_c)	25%	**74.0**	Max class posterior margin (Confidence: 78.3%)
3. Boundary Margin (S_m)	20%	**100.0**	Separation distance from referral threshold (|P - 0.30| = 0.700)
4. Biomarker Completeness (S_comp)	15%	**100.0**	Integrity across quality, vessels, anatomy, lesions & CNN pipelines
5. Explainability Concordance (S_agree)	15%	**97.5**	CAM activation and physical lesion overlap (Dice = 0.214)

COMPOSITE RELIABILITY: 90.0 / 100 | TIER: HIGH | SAFETY GATE: PASSED (tau_accept = 60.0)

VISUAL INTERPRETABILITY ATLAS — PART 1: PRIMARY IMAGING & ANATOMY

1. Original Retinal Fundus Photograph

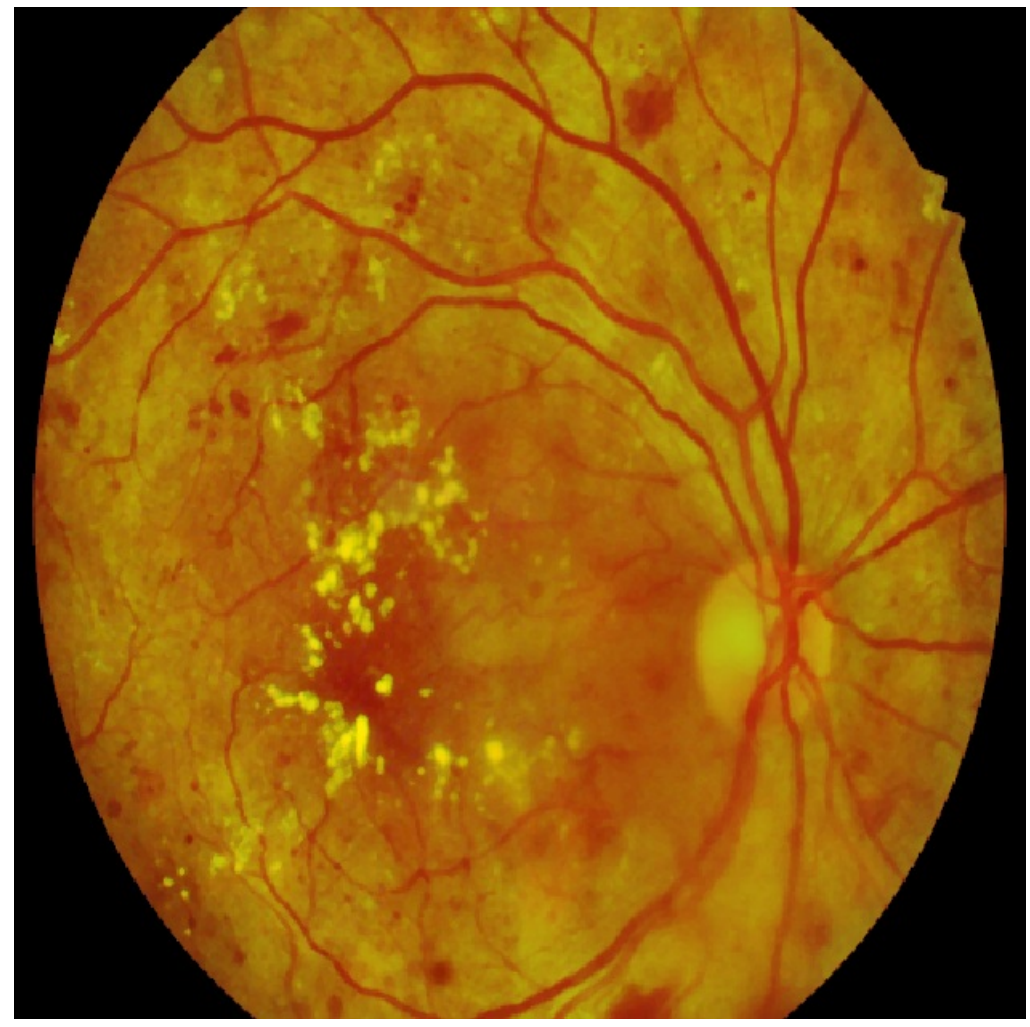


True color 45-degree retinal fundus image as acquired from the camera.

Legend: RGB Fundus Image (Baseline Acquisition)

Method: Direct Acquisition (Verified)

2. Preprocessed & CLAHE Contrast Enhanced



Normalized illumination with shading correction and CLAHE contrast enhancement.

Legend: Contrast Enhanced (Green Channel Priority)

Method: Preprocessing Pipeline (Active)

3. Retinal Anatomical Landmarks

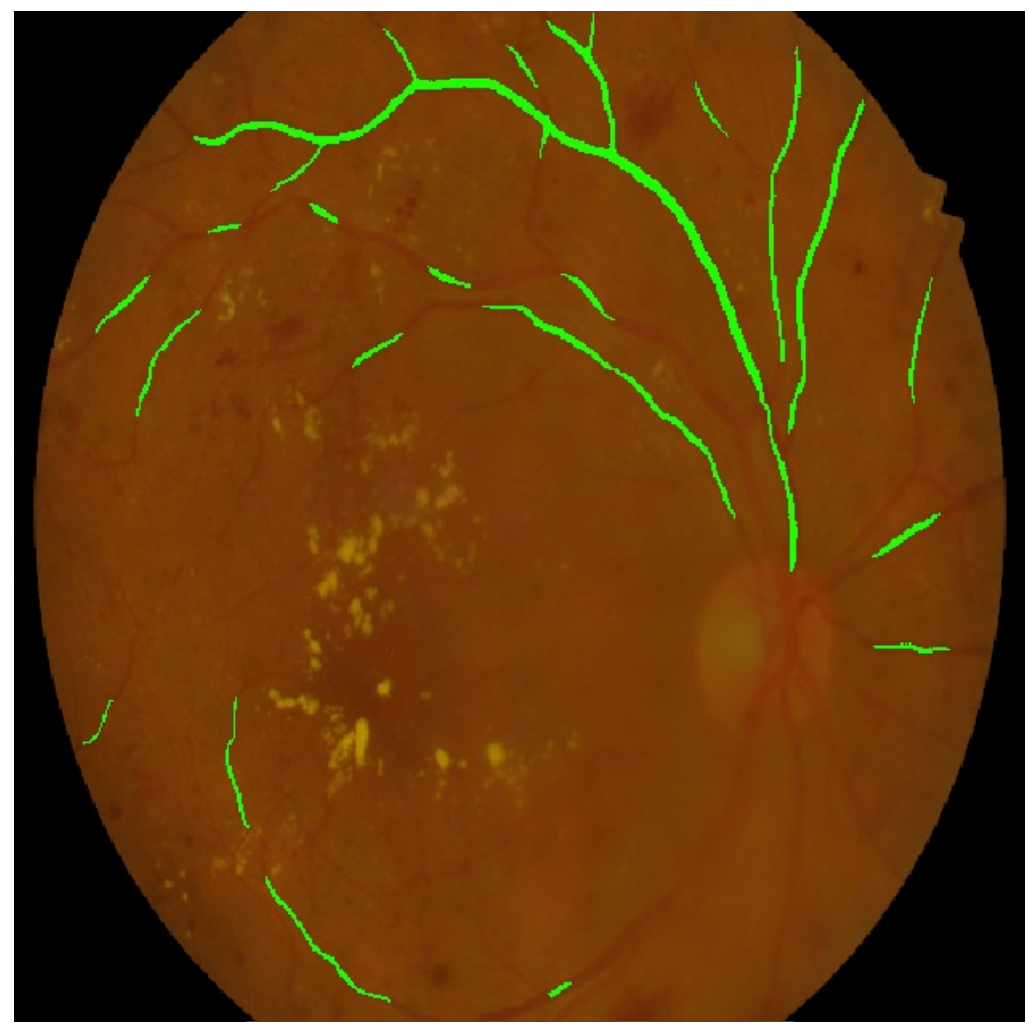


Cyan circle: Optic Disc boundary. Amber ring: Macular zone (2.5x OD radius).

Legend: Cyan: Optic Disc | Amber: Macula Zone

Method: Geometric Morphometry (Calibrated)

4. Retinal Vessel Network Architecture



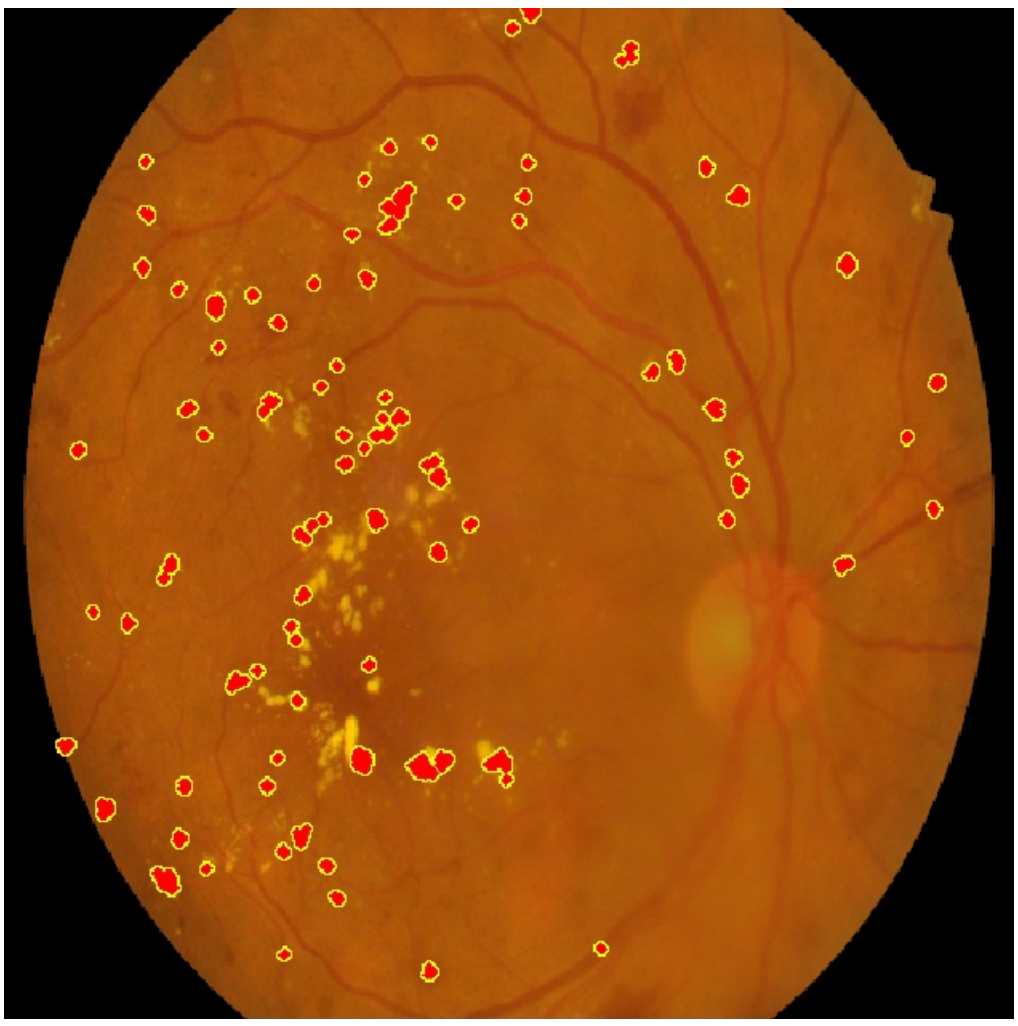
Deep U-Net vessel network segmentation overlaid on normalized fundus.

Legend: Green: Segmented Retinal Vessel Network

Method: Deep U-Net Model (DRIVE/IDRiD)

VISUAL INTERPRETABILITY ATLAS — PART 2: LESION BIOMARKER OVERLAYS

5. Microaneurysms Detection (MAs)

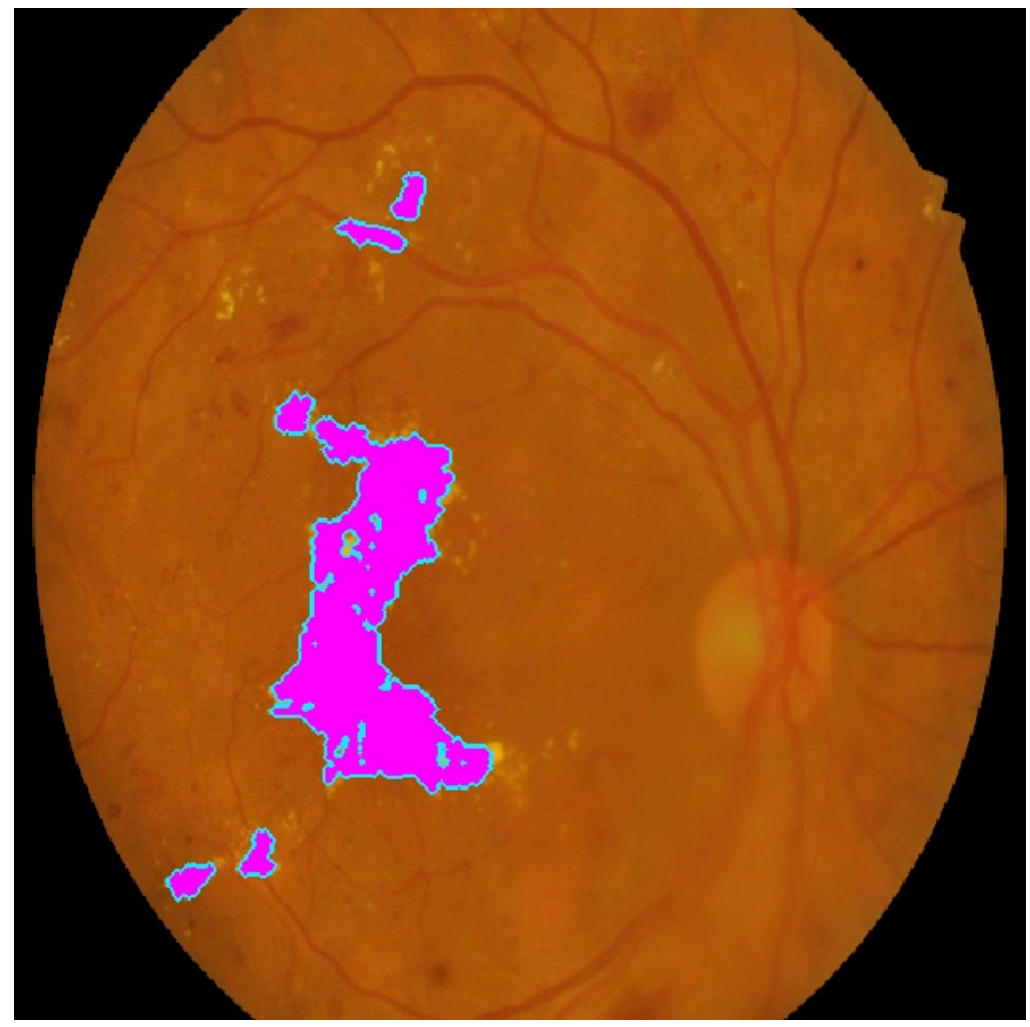


Red bounding markers: Detected microaneurysm candidates.

Legend: Red center with yellow target ring: MAs

Method: Multi-Scale Morphological Filter

6. Intraretinal Hemorrhages (HEs)

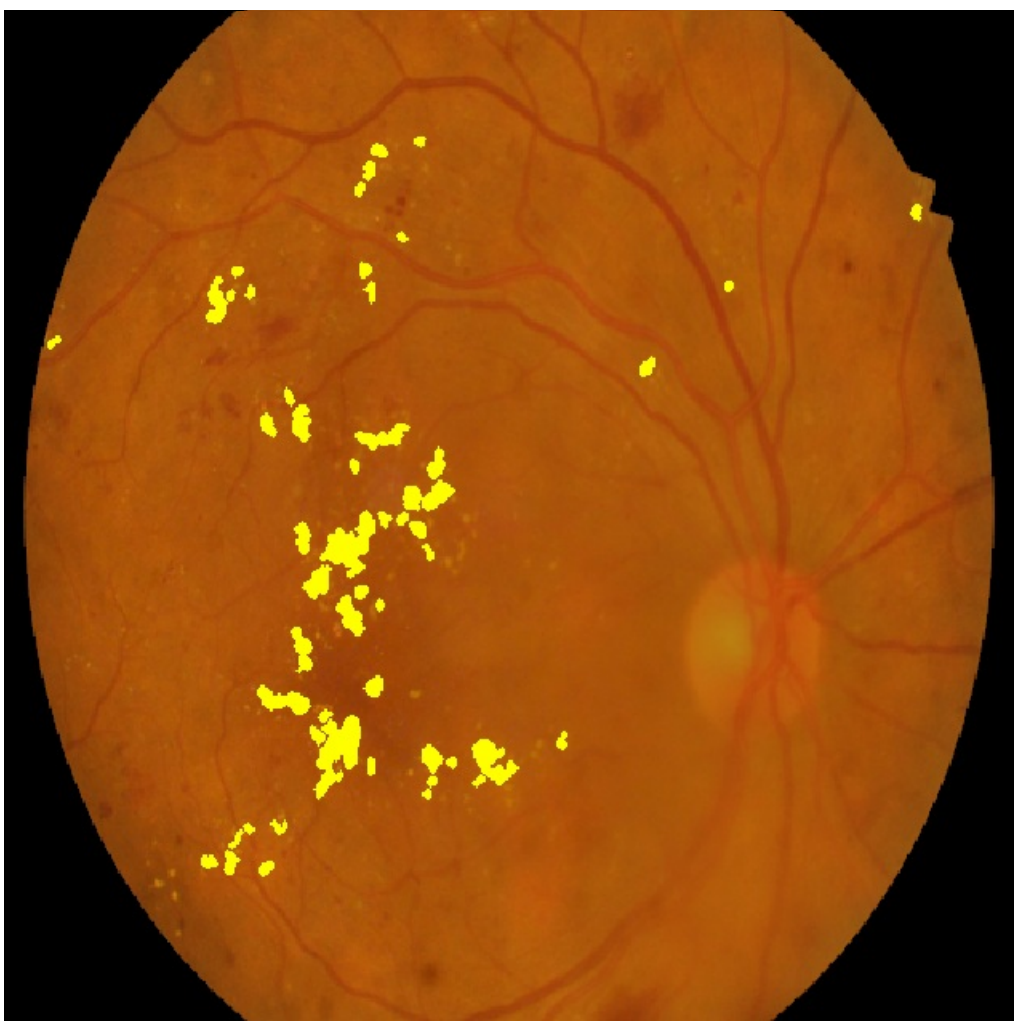


Magenta regions: Detected intraretinal hemorrhage boundaries.

Legend: Magenta fill with cyan delineation: HEs

Method: Multi-Scale Thresholding

7. Hard Exudates & DME Risk Map

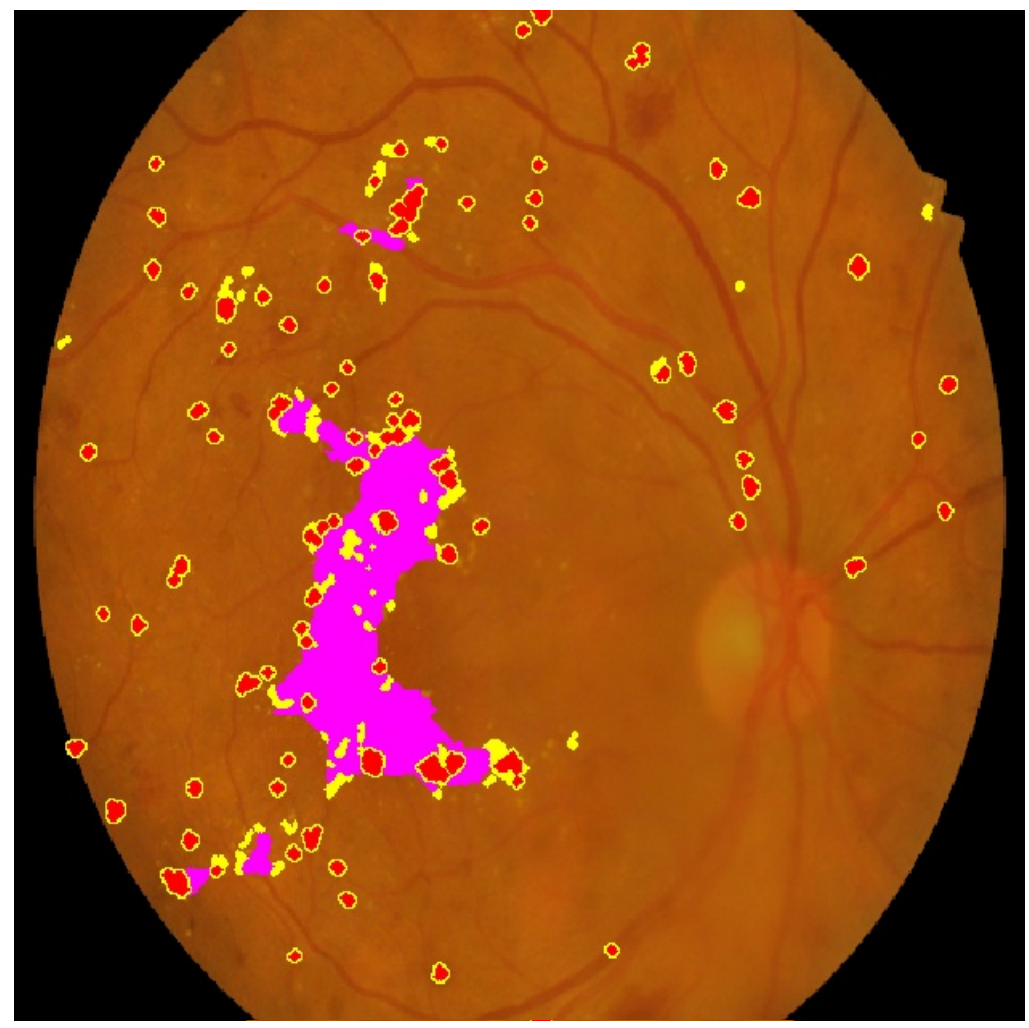


Yellow regions: Hard lipid exudates. Circles: Foveal proximity risk zones.

Legend: Yellow: Exudates | Amber: Threat Zone

Method: Morphological & Macular Proximity

8. Multi-Lesion Clinical Composite



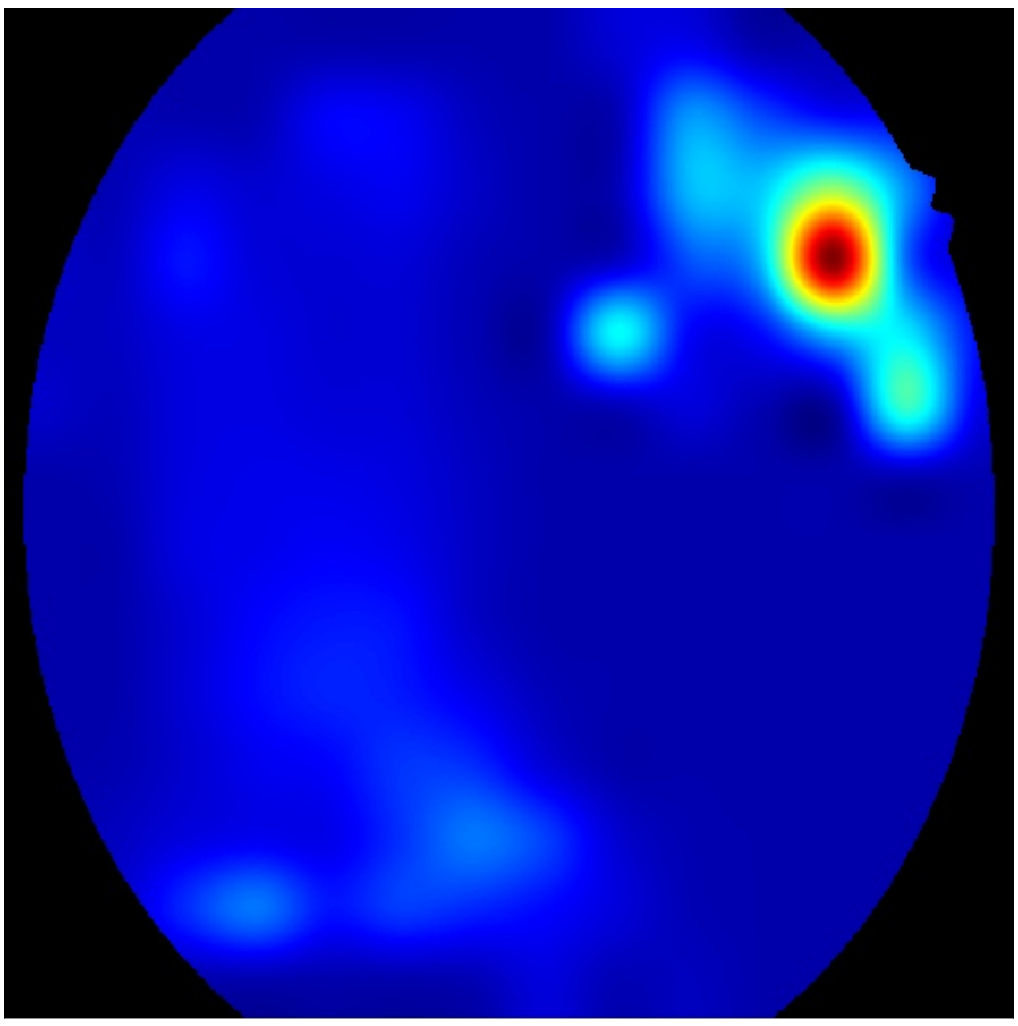
Combined clinical biomarker overlay: Red (MA), Magenta (HE), Yellow (EX).

Legend: Yellow: EX | Magenta: HE | Red/Yellow: MA

Method: Multi-Biomarker Composite Map

VISUAL INTERPRETABILITY ATLAS — PART 3 & PROTOCOL SUMMARY

9. Grad-CAM Deep Activation Heatmap

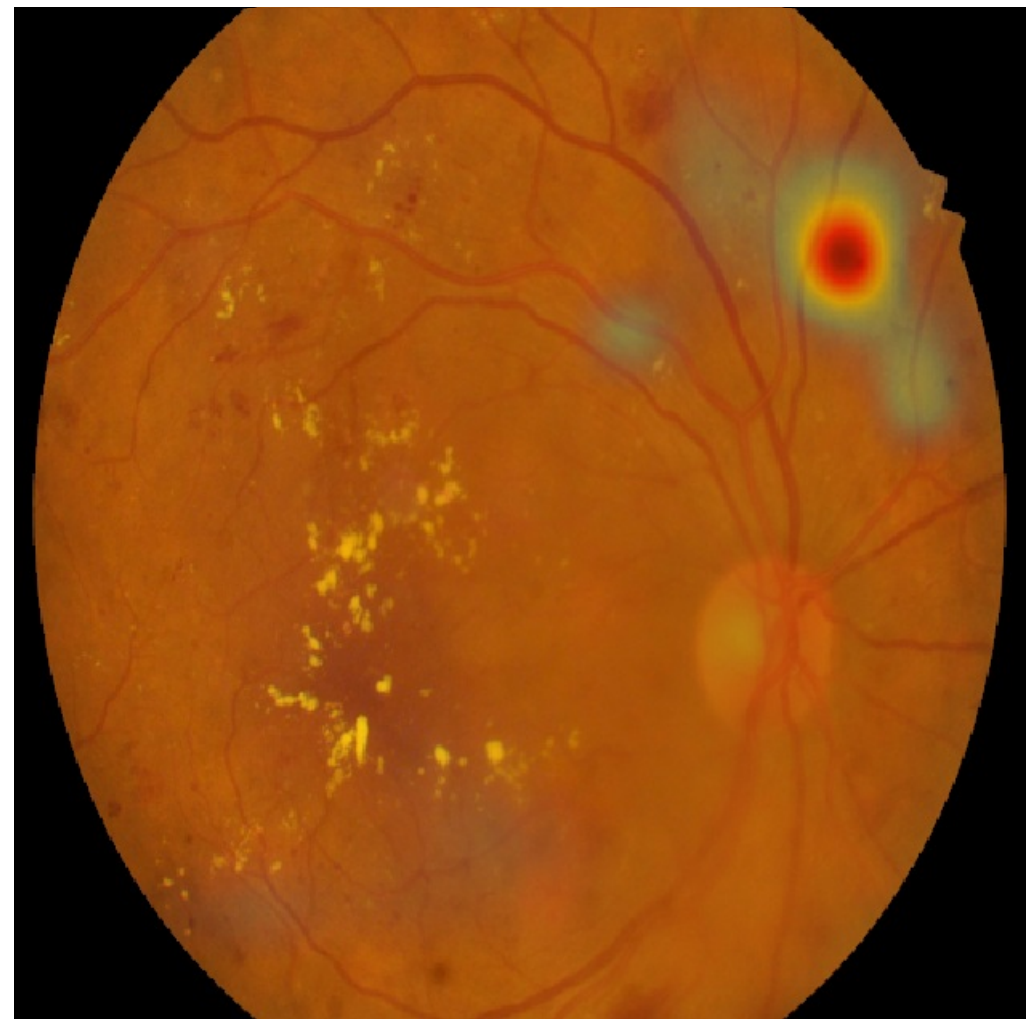


Class activation heatmap from SqueezeNet final convolutional layer.

Legend: Jet Colormap (Red=High attention)

Method: Deep Grad-CAM (SqueezeNet V3)

10. Grad-CAM Clinical Explainability Blend



Alpha-blended Grad-CAM attention overlay providing spatial interpretability.

Legend: Weighted Alpha-Blend (Heatmap + Fundus)

Method: Spatial CAM-Lesion Dice Alignment

CLINICAL TRIAGE & TELE-OPHTHALMOLOGY WORKFLOW SUMMARY

PRIMARY HEALTH CENTER PROTOCOL RECOMMENDATIONS:

- Examination Outcome: REFERABLE DR (Grade 3)
- System Triage Disposition: ACCEPT SCREENING
- Autonomous Decision Readiness: ACCEPTED (Meets high reliability threshold)
- Recommended Clinic Action: High-confidence referable DR detected. Schedule clinical examination with ophthalmologist.
- Recommended Referral Timeline: 2 to 4 weeks (Ophthalmology Clinic Referral)

NEURAL ATTENTION & SPATIAL CONCORDANCE:

- CAM-Lesion Concordance (Dice): 0.214 (Spatial alignment between neural attention and physical lesions)
- Clinical Interpretability: Network focus corresponds to pathognomonic retinal lesion distributions.

DOCUMENTATION & AUDIT TRAIL:

- Software Engine Version: SHINRA Diagnostic Framework Release V2 (Development Frozen)
- Calibration Checkpoint: Platt Parameters a = 1.3106, b = -1.1412 (Fitted on IDRiD 330-train set)
- Operating Threshold: Referral tau_ref = 0.30 | Acceptance tau_accept = 60.0
- Destination Report File: D:/SHINRA/results/reports/test_regression_report.pdf

CLINICAL REVIEWER VERIFICATION & SIGN-OFF:

Reviewer Name: _____ Medical Reg. No: _____

Review Date: _____ Signature: _____